

Andreas Faust

Publications

A. Faust, S. Nitzsche, J. Becker. *Efficient Respiratory Sound Denoising with a Distilled Spiking Network.* IEEE BioCAS 2026, Incheon. Winning entry of the BioCAS 2026 Grand Challenge, presented by the author.

S. Nitzsche, **A. Faust**, J. Krausse, S. Sinn, B. Pachideh, J. Becker. *Beyond Leaky Integrate-and-Fire: Evolving Spiking Neuron Models for Efficient Machine Learning.* IEEE BioCAS 2026, Incheon.

A. Faust, S. Nitzsche, J. Becker. *Zero-Inflated Gaussian Distributions Enable Parameter-Space Sparsity in Estimation-of-Distribution Algorithms.* Preprint, arXiv:2606.19369, 2026.

S. Nitzsche, A. Ionita, **A. Faust**, B. Ionescu, J. Becker. *An Optimization Framework for Automated Assessment of Biological Plausibility of Spiking Neurons.* ECML PKDD 2025 Workshops, Springer CCIS vol. 2840, pp. 364–379, 2026.

Education

University of Freiburg

Master of Science in Computer Science
Apr 2025 - Today

Karlsruhe Institute of Technology

Bachelor of Science in Computer Science
Oct 2017 - Apr 2024

Balluff GmbH

Journeyman of Mechatronics
Oct 2014 - Jan 2017
• PLC, CNC Machining, Pneumatics, Electronics

Skills

Programming Languages: C, C++, C#, Python, Java, glsl
Areas of Expertise: Deep Learning, Data Science, [Convolutional-, Recurrent-, Graph-, Spiking-] Neural Networks, Embedded ML, Genetic Algorithms, Computer Vision, Embedded C, Parallel Computing, Electronics
Technologies: Git, Gitlab, GitHub, Docker, Ubuntu, SSH, REST APIs, OpenGL, Pytorch, Tensorflow, OpenCV, PyBind11, Optuna, Tensorboard, Weights and Biases, Matplotlib, Seaborn, Pandas, Jupyter
Languages: Fluent in English and German

Professional Experience

FZI - Research Center for Computer Science

Research Assistant
May 2024 - Jun 2024, Dec 2024 - Today
• Evolutionary optimization of spiking neuron models, continuing my bachelor's thesis work
• Participation in IEEE BioCAS 2026 Grand Challenge on respiratory sound denoising
• First-author publication on sparse estimation-of-distribution algorithms

AMAI GmbH

Junior Machine Learning Engineer
Sep 2024 - Nov 2024
• Consulting companies on implementing AI solutions

Karlsruhe Institute of Technology - IAR-ALR Institute

Student Research Assistant
May 2022 - Oct 2022
• Implementation and testing of new features for a Message Passing Graph Neural Network Python library

Apic.ai

Software Developer
Mar 2019 - Dec 2020
• Developed a Multi-Object-Tracking library for tracking up to 200 bees simultaneously on a video stream
• Optimized performance-critical code to run inference on embedded ML hardware
• Designed and managed neural networks & training data

Additional Activities

Volunteer Work
May 2017 - Sep 2017 &
Sep 2022 - Oct 2022 &
Jul 2024 - Sep 2024
Archelon NGO, Greece
• Protecting sea turtle nests
• Raising public awareness through presentations in English and German